An angle-dependent estimation of CT x-ray spectrum from rotational transmission measurements

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Purpose: Computed tomography (CT) performance as well as dose and image quality is directly affected by the x-ray spectrum. However, the current assessment approaches of the CT x-ray spectrum require costly measurement equipment and complicated operational procedures, and are often limited to the spectrum corresponding to the center of rotation. In order to address these limitations, the authors propose an angle-dependent estimation technique, where the incident spectra across a wide range of angular trajectories can be estimated accurately with only a single phantom and a single axial scan in the absence of the knowledge of the bowtie filter.
Methods: The proposed technique uses a uniform cylindrical phantom, made of ultra-high-molecular-weight polyethylene and positioned in an off-centered geometry. The projection data acquired with an axial scan have a twofold purpose. First, they serve as a reflection of the transmission measurements across different angular trajectories. Second, they are used to reconstruct the cross sectional image of the phantom, which is then utilized to compute the intersection length of each transmission measurement. With each CT detector element recording a range of transmission measurements for a single angular trajectory, the spectrum is estimated for that trajectory. A data conditioning procedure is used to combine information from hundreds of collected transmission measurements to accelerate the estimation speed, to reduce noise, and to improve estimation stability. The proposed spectral estimation technique was validated experimentally using a clinical scanner (Somatom Definition Flash, Siemens Healthcare, Germany) with spectra provided by the manufacturer serving as the comparison standard. Results obtained with the proposed technique were compared against those obtained from a second conventional transmission measurement technique with two materials (i.e., Cu and Al). After validation, the proposed technique was applied to measure spectra from the clinical system across a range of angular trajectories [$-15°$, $15°$] and spectrum settings (80, 100, 120, 140 kVp).
Results: At 140 kVp, the proposed technique was comparable to the conventional technique in terms of the mean energy difference (MED, -0.29 keV) and the normalized root mean square difference (NRMSD, 0.84%) from the comparison standard compared to 0.64 keV and 1.56%, respectively, with the conventional technique. The average absolute MEDs and NRMSDs across kVp settings and angular trajectories were less than 0.61 keV and 3.41%, respectively, which indicates a high level of estimation accuracy and stability.
Conclusions: An angle-dependent estimation technique of CT x-ray spectra from rotational transmission measurements was proposed. Compared with the conventional technique, the proposed method simplifies the measurement procedures and enables incident spectral estimation for a wide range of angular trajectories. The proposed technique is suitable for rigorous research objectives as well as routine clinical quality control procedures.
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Key words: CT, spectral estimation, transmission measurements, x-ray spectrum

1. INTRODUCTION

With the improvements of the diagnostic imaging technologies, there are increasing demands for the accurate knowledge of the x-ray spectrum produced by a particular x-ray source. For example, in dose and risk estimations for computed tomography (CT),[1,2] the x-ray spectrum is used in Monte Carlo simulations to determine the energy deposition in a patient body. In polyenergetic reconstruction[3,4] or dual energy reconstruction,[5] the x-ray spectrum can be used as *a priori* information to reduce beam hardening artifacts and to implement quantitative imaging approaches. It is also desirable to measure the x-ray spectrum for quality assurance purposes, as the spectrum of an x-ray source can drift from the expected values due to repeated usage.[6]

Current spectral estimation techniques can be roughly categorized into three approaches: (1) computer simulation, (2) spectroscopic measurements, and (3) transmission-based measurements.

Computer-simulation-based approaches[7–9] aim to estimate generalized spectra based on basic physical parameters, such as the kinetic energy of the incident electrons, the density and the attenuation coefficient of the target material, and the target angle. The characteristic radiation is added to the Bremsstrahlung continuum according to empirical relationships.[8] These approaches can model fundamental mechanics of the x-ray formation process, but are generally unable to fully account for the hardware nuances of individual x-ray systems (e.g., the used condition of the x-ray tube and the actual attenuation properties of the inherent filtrations and the bowtie filters).

Spectroscopic measurement approaches[10–15] assess the spectrum from a particular x-ray source using energy-resolved, photon-counting x-ray detectors (e.g., CdTe detectors[12–14] or CdZnTe detectors[15]). While these approaches generally give a more representative estimation of the actual spectrum (considering the limitation implied by the experimental uncertainties), they are not suitable for routine measurements because they require expensive experimental tools (e.g., dedicated detectors) and delicate experimental settings. For example, in the preparation phase, the detector must be calibrated with radioactive elements (e.g., ^{241}Am and ^{133}Ba) and the whole experimental setup needs to be aligned precisely. In addition, the spectrum correction procedure[15] requires a detailed understanding of the detector properties, such that Monte Carlo simulation can be used to compute the response functions for each detector channel, with which the distorted spectrum can be corrected by stripping methods.[15,16]

A more practical way to the spectral estimation is called transmission-measurement-based approaches, which use transmission measurements of phantoms of known dimensions and compositions.[6,17–19] These approaches are relatively simple and have been shown to provide satisfactory results for various energy ranges of x-ray spectra.[6,17] However, both the phantom preparation and the measurement process of these approaches can be time consuming and potentially error prone. For instance, in the phantom preparation, two or more materials have to be used and each material has to be carefully shaped and stacked-up into different thicknesses. In the measurement process, all phantoms have to be sequentially exposed to x rays.

One of the major drawbacks of these approaches is that, without the knowledge of the bowtie filter (i.e., shape, composition, and position), it is difficult to derive the incident spectra across the entire CT scan field of view (SFOV). Moreover, to obtain the transmission measurements, these approaches often have to be implemented in a special service acquisition mode, where the gantry is stationary.[18] This operational mode is not always available to practicing medical physicists and requires special service personnel from the manufacturer to access the scanner.

In this work, we propose a new transmission-measurement-based approach for clinical CT scanners. Using a single cylindrical phantom and a single axial scan, incident spectra across a wide range of angular trajectories can be estimated without the knowledge of the bowtie filter characteristics. The method still requires a certain level of access to the raw data from the manufacturer. However, the measurements can be made without having the gantry in the stationary mode. In this paper, we first review the mathematical model of the transmission-measurement-based approaches in Sec. 2.A and then introduce the proposed off-centered scanning geometry and the data conditioning procedure in Secs. 2.B and 2.C, respectively. This paper further details the experimental validation of the proposed technique in Sec. 2.D.

2. METHODS

2.A. Mathematical model

The transmission measurements can be expressed as an integral equation as

$$Y(L) = \int_E I(E) e^{-\mu(E)L} dE = \int_E N(E)\eta(E) E e^{-\mu(E)L} dE, \quad (1)$$

where $Y(L)$ is the transmission measurement, $I(E)$ is the photon intensity at energy E, $\mu(E)$ is the attenuation coefficient of the phantom, and L is the x-ray intersection length through the phantom. As a conventional x-ray imaging CT system has energy-integrating detectors, $I(E)$ can be further broken into three components: the incident spectrum $N(E)$ (the objective of the characterization), the detector response $\eta(E)$, and the photon energy E. In this work, $\int I(E)dE$ is normalized to one, so the transmission measurement $Y(L)$ is equivalent to the transmission ratio.

Generally, the measurement expressed in Eq. (1) follows a compound Poisson distribution.[20] Previously developed statistical image reconstruction methods have used the assumption of simple Poisson statistics[3,21,22] with acceptable results. In this work, we adopted the latter assumption, i.e., each measurement $Y(L)$ is assumed to follow a simple Poisson distribution with the mean intensity of $\int_E I(E)e^{-\mu(E)L}dE$ as

$$Y(L) = \text{Poisson}\left\{\int_E I(E)e^{-\mu(E)L}dE\right\}, \quad (2)$$

or

$$Y_m = \text{Poisson}\left\{\sum_{s=1}^{N_S} A_{m,s}I_s\right\}, \quad m = 1, \ldots, N_M, \quad (3)$$

in the discretization form, where N_M is the total number of transmission measurements, N_S is the number of samplings of the spectrum, $A_{m,\,s}$ is the system matrix element computed from the phantom attenuation, the phantom thickness, and the size of the discretized spectral interval ΔE using the relation

$$A_{m,s} = e^{-\mu_s L_m}\Delta E. \quad (4)$$

Based on the simple Poisson assumption, the probability of obtaining Y_m with the photon intensity spectrum I_s can be computed as

$$P(\mathbf{I}|\mathbf{Y},\mathbf{A}) = \prod_m \frac{\left(\sum_s A_{m,s}I_s\right)^{Y_m}}{(Y_m)!}e^{-\sum_s A_{m,s}I_s}, \quad (5)$$

where $\boldsymbol{I} = \{I_1, \ldots, I_{N_S}\}^T$, $\boldsymbol{Y} = \{Y_1, \ldots, Y_{N_M}\}^T$, and $\boldsymbol{A} = \{A_{m,s}\} \in R^{N_M \times N_S}$. In order to estimate I_s, one possible approach is to maximize the log-likelihood objective function of Eq. (5) through

$$\mathbf{I} = \arg\left\{\max_{\mathbf{A},\mathbf{Y}} \ln P(\mathbf{I}|\mathbf{A},\mathbf{z})\right\}$$
$$= \arg\left\{\max_{\mathbf{A},\mathbf{Y}} \sum_m \left[Y_m \ln\left(\sum_s A_{m,s}I_s\right) - \sum_s A_{m,s}I_s\right] + c\right\}, \quad (6)$$

where c is a constant. As a logarithmic function is concave, based on Jensen's inequality, Eq. (6) can be equivalently converted to

$$\mathbf{I} = \arg\left\{\max_{\mathbf{A},\mathbf{Y}} \sum_{m,s}\left[Y_m \ln(A_{m,s}I_s) - A_{m,s}I_s\right] + c\right\}. \quad (7)$$

Since this objective function is not quadratic, the Expectation-Maximization (EM) approach[23,24] is applied to solve this maximization problem. The resulting iteratively multiplicative update form is

$$I_s^{(k)} = \frac{I_s^{(k-1)}}{\sum_m A_{m,s}} \sum_m \frac{A_{m,s}Y_m}{\sum_{s'} A_{m,s'}I_{s'}^{(k-1)}}, \quad (8)$$

where k indicates the kth iteration. For the EM method, the main features of the spectrum (such as the characteristic x rays of the x-ray tube and the K-edges of the detector) must be reflected in the initial guess ($I_s^{(0)}$).[6,18] After each iteration, the sum of $\sum I_s^{(k)}$ is normalized to one. The iteration can be terminated when the averaged error is smaller than a threshold T as

$$\frac{1}{N_M}\sum_{m=1}^{N_M}\left|Y_m - \sum_{s=1}^{N_S} A_{m,s}I_s^{(k)}\right| \le T. \quad (9)$$

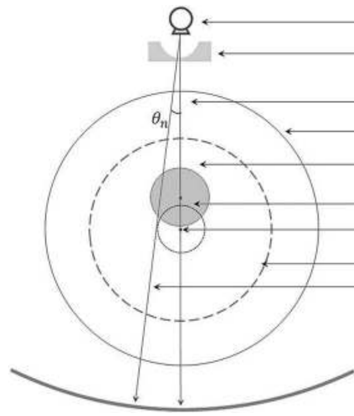

FIG. 1. Off-centered acquisition geometry for the angle-dependent estimation technique of x-ray spectrum.

Usually, the iterative equation [Eq. (8)] converges within about 3.0×10^3 iterations for $T = 10^{-5}$. The computational time is less than 2 s using Matlab running on a desktop computer (3 GB RAM and 2.5 GHz CPU). With I_s, the incident spectrum N_s can be obtained by dividing the detector response η_s and the energy E_s as

$$N_s = \frac{I_s}{\eta_s E_s}. \quad (10)$$

2.B. Acquisition geometry

Because of the bowtie filter, the incident spectrum $N(E)$ gradually changes as a function of beam angle. In order to fast and accurately sample the transmission measurements across a wide range of angular trajectories, we propose an off-centered acquisition geometry as shown in Fig. 1.

Using a cylindrical shaped phantom and targeting the transmission measurements of the nth detector element, the x-ray beam from the x-ray source to the nth detector element is always tangent to a circle (dotted line in Fig. 1) with its center located at the center of the gantry rotation. Based on this circle, one can trace the position of this x-ray beam and determine the corresponding intersection length during the rotation of the gantry (Fig. 2). Starting from position 1 [Fig. 2(a)], the intersection length gradually increases to the maximum (i.e., the diameter of the phantom) at position 2 [Fig. 2(b)], decreases to zero at position 3 [Fig. 2(c)], and then starts to increase again from zero value at position 4 [Fig. 2(d)]. Based on this off-centered geometry, the detectors from different angles can quasicontinuously record a series of transmission measurements with a large range of intersection lengths.

In order to make sure the ranges of the intersection lengths of the central detector elements can always start from zero, the phantom should be kept completely off from the center of rotation (e.g., position 1 in Fig. 3). With the increase of the distance between the phantom and the rotation center, the phantom can cover a larger range of angular trajectories. The furthest location to position the phantom is position 2 in Fig. 3, where phantom is tangent to the SFOV. When the detector is rotated to the closest location to the phantom

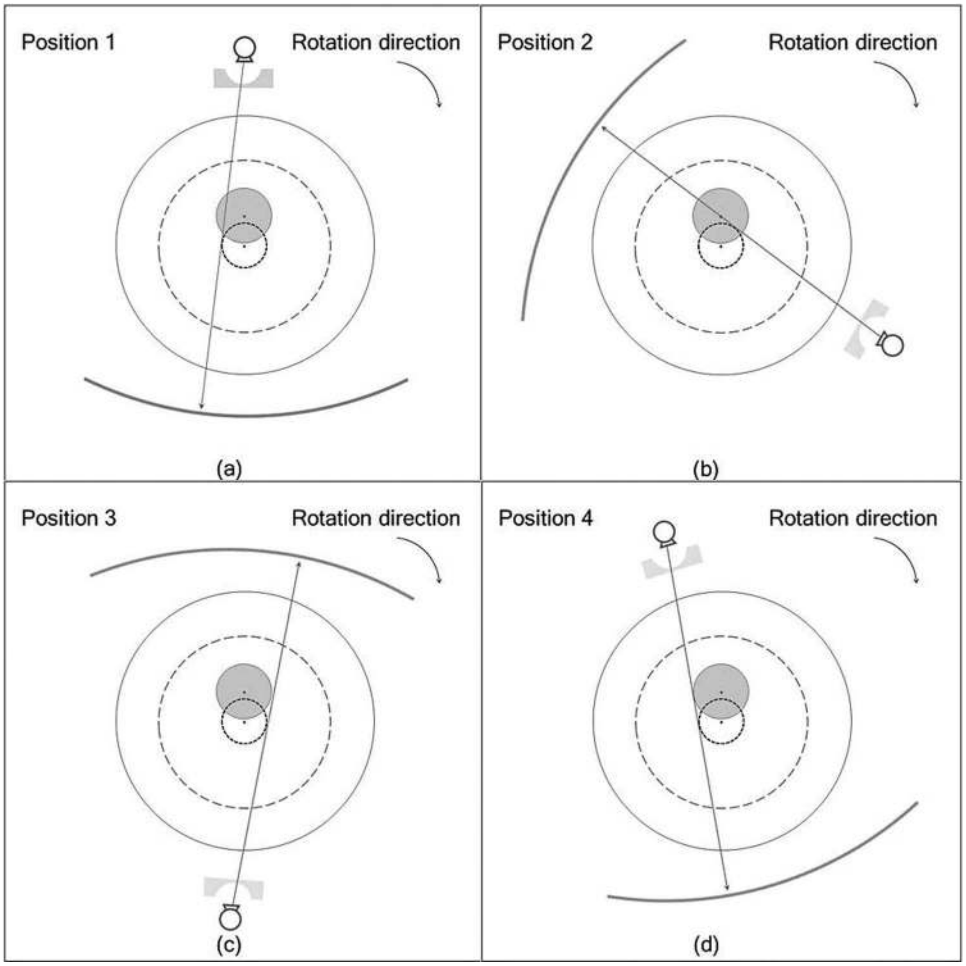

FIG. 2. (a)–(d) Schematic diagram of the change of the attenuation path of the x-ray beam from the x-ray source to the *n*th detector element during the CT scan in axial mode.

during the CT scan, scatter radiation can increase proportionally (see our Monte Carlo simulation results in Sec. 2.D). As a scanned object is usually positioned in the center, the built-in model-based correction techniques[25,26] can effectively reduce the scatter radiation via tabulated tables. Therefore, it would be advantageous for the phantom to be positioned only slightly off-centered as in position 1 (Fig. 3), so that the scatter radiation pattern is more similar to that of the centered phantom and the scatter radiation across all projections can be readily corrected by the scanner's built-in scatter correction techniques. Positioning the phantom in position 2 provides a wider sampling of the angular trajectories but a larger impact of scatter radiation on the measurements. Note that phantom located beyond position 2 (e.g., position 3 in Fig. 3) will further result in truncated projections and the cross sectional image cannot be accurately reconstructed for the derivation of the intersection lengths using conventional FBP method. Dedicated reconstruction algorithms for truncated projections have to be employed to extend the CT SFOV (Refs. 27 and 28) for this case.

As the phantom is positioned within the SFOV and the CT scanner is operated in axial mode with the x rays continuously on for a full rotation, the collected transmission data can be used to reconstruct the cross-sectional image of the scanned phantom. The voxel values that are larger than half of the attenuation of the scanned material can be set to unit one to facilitate deriving the intersection lengths via the forward ray tracing procedure. If the shape of the phantom is convex, a

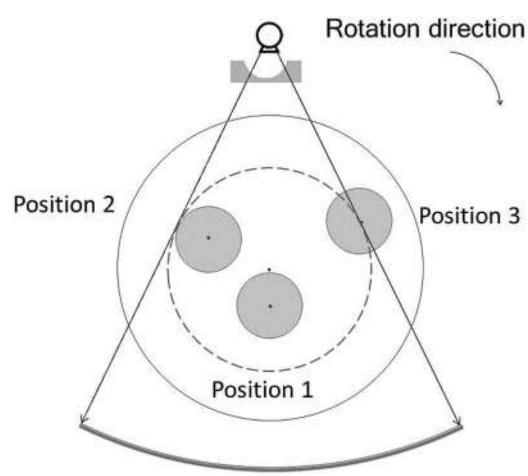

FIG. 3. Possible off-centered positions. At position 1, the distance between the center of the phantom and the rotation center is slightly larger than the radius of the phantom. At position 2, the phantom is tangent to the SFOV. At position 3, a part of the phantom is positioned outside of the SFOV.

conventional backward ray tracing technique can be used to delineate the border of the phantom instead of using image reconstruction.

The phantom should be made of a material with moderate attenuation properties. Materials with large attenuations (such as metals) could be small in size, but the accuracy of the intersection lengths derived from the reconstructed image will suffer large errors due to the limited resolution of the CT detectors (0.6×0.6 mm^2 at isocenter). Materials with small attenuations provide for a larger phantom enabling smaller relative errors in intersection length measurement, but can lead to increased scatter radiation to the detectors. Plastic materials, such as ultra-high-molecular-weight polyethylene [UHMWPE; chemical formula: $(C_2H_4)_nH_2$, density: 0.937 g/cm^3], have moderate attenuations. In this work, a 160 mm diameter cylindrical phantom made of polyethylene was used to reduce transmission to 0.1 for the maximum path length through the phantom for a typical 140 kVp beam.

2.C. Data conditioning procedure

The off-centered acquisition geometry can collect hundreds of transmission measurements across a wide range of angular trajectories, which provides abundant experimental data. However, the large amount of data can make the estimation process time consuming, while the data are further contaminated by various sources of noise (such as quantum noise, instrumentation noise, or stochastic variations due to the phantom and the experimental condition). In order to accelerate spectral estimation speed, to reduce noise, and to improve estimation stability, the data are preprocessed before spectral estimation.

Generally, the transmission measurements can be fitted with polynomial functions. Equation (1) can be expanded in a Taylor series as

$$Y(L) = \int_E I(E) e^{-\mu(E)L} dE = \sum_k c_k L^k, \tag{11}$$

where

$$c_k = \int_E \frac{I(E)\,[-\mu(E)]^k}{k!} dE. \tag{12}$$

Because the intersection length L can be very large (e.g., 160 mm), it is necessary to ensure that the Taylor series converges with a suitable truncation degree. In order to make this determination, the absolute value of the nth term ($|c_k L^k|$, where $L = 160$ mm) of Eq. (11) is plotted in Fig. 4 as a function of polynomial degree (k) for different spectra. In this work, we used a polynomial function with a truncation degree of 11 to maintain errors to be less than 10^{-2}.

One boundary condition that should be imposed to the polynomial function is that the transmission measurements Y_m should always be 1 when x-ray beam is not attenuated ($L = 0$). Because of the limited resolution of the detector and various sources of noise, when an x-ray path approaches the edge of the phantom, the derived intersection length might

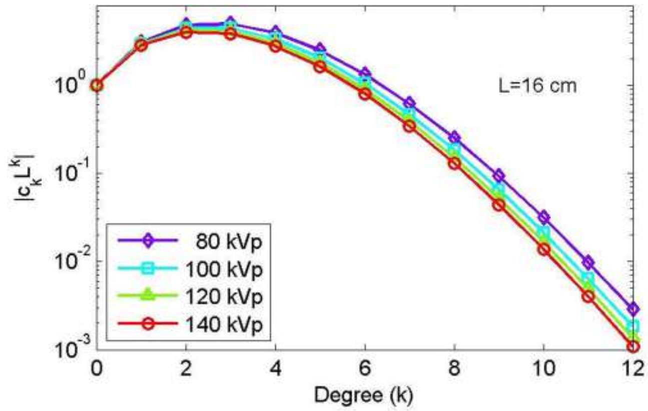

FIG. 4. Absolute value of the kth term ($|c_k L^k|$, where $L = 160$ mm) of the Taylor series in Eq. (11) as a function of polynomial degree (k) for different spectra.

change dramatically and the fitted function may not necessarily satisfy this boundary condition. In order to force the polynomial functions through this fixed point, c_0 in Eq. (11) is set to unity. In matrix notation, the equation for a polynomial fit is given by

$$\mathbf{Y} - \mathbf{1} = \mathbf{L}\mathbf{c}, \tag{13}$$

where $\mathbf{1} = [1, \ldots, 1]^T \in R^{N_M \times 1}$, $\boldsymbol{L}$ is the Vandermonde matrix generated by the L_m, i.e.,

$$\mathbf{L} = \begin{bmatrix} L_1^1 & \cdots & L_1^{N_K} \\ \vdots & \ddots & \vdots \\ L_{N_M}^1 & \cdots & L_{N_M}^{N_K} \end{bmatrix}, \tag{14}$$

and $\boldsymbol{c} = [c_1, \ldots, c_{N_K}]^T \in R^{N_K \times 1}$ is the vector of unknown polynomial coefficients. In this work, the linear equation system of Eq. (13) is solved using a fixed point least square method as

$$\mathbf{c} = (\mathbf{L^T L})^{-1}\mathbf{L^T}(\mathbf{Y} - \mathbf{1}). \tag{15}$$

With the fitted polynomial function for each angular trajectory, we resampled 30 measurements equally distributed between $L = 0$ and the measured maximum intersection length for the spectral estimation.

2.D. Experimental validation

The proposed spectral estimation technique was validated on a CT scanner (Somatom Definition Flash, Siemens Healthcare, Germany). Details of the prebowtie spectra (140, 120, 100, and 80 kVp), the bowtie filter, and the detector absorption ratios were provided from the manufacturer. These spectra were measured with a spectroscopic x-ray detector using a Compton scattering method.[10,11,13] The bowtie filter was used to derive the incident spectra along different angular trajectories from manufacturer-provided prebowtie spectra. The detector absorption ratios were used to convert the estimated photon intensities to incident spectra [Eq. (10)]. The CT geometric information and the scan parameters used are specified in Table I.

TABLE I. CT geometric information and scan parameters used in this work.

Parameter name	Value
Source-to-detector distance	1085.6 mm
Source-to-object distance	595.0 mm
Gantry opening	780 mm
Number of detector columns	736
Fan angle	49.95°
Detector size at isocenter	0.60×0.60 mm^2
Detector collimation	2×1 mm
Bowtie filter	w3
Tube current	500 mAs
Exposure time	1.0 s
Number of projections	1152 projections for the proposed technique

In order to reduce the potential influence of the quantum noise, a high mAs setting (i.e., 500 mAs) was used. The obtained two-row data were then averaged with respect to each column to further suppress the quantum noise.

A comparative experiment was conducted between the conventional technique and the proposed technique. For the conventional technique,[18] two materials (i.e., Al and Cu) were selected. For each material, nine metal filters of different thicknesses were used. The thicknesses of the metal filters were selected to ensure uniformly distributed transmission measurements within the [0.0, 1.0] range (Table II). Actual thicknesses of the metal filters were measured with a caliper. For this technique, the scanner was operated in the service mode,[18] i.e., both the x-ray tube and the detector remained stationary. The metal filters were positioned in a proximate location to the x-ray tube to minimize the scatter radiation and the transmission measurements through the metal filters were acquired sequentially along the central axis of the beam.

For the new technique, a cylindrical phantom[29] with a 160 mm diameter was used, which was made of uniform ultra-high-molecular-weight polyethylene. The experimental setup is shown in Fig. 5. First, the scanning section of this phantom was extended beyond the patient table to avoid the influence of the patient bed [Fig. 5(a)]. Second, this phantom was positioned off center in the SFOV by adjusting vertically the patient bed [Fig. 5(b)]. With the proposed technique, the scanner was able to be operated in the axial mode. After measurement, raw data were exported from the scanner to an external computer, from which the transmission data were extracted for spectral estimation via a manufacturer-provided program. The beam-hardening corrections were excluded from the transmission data.

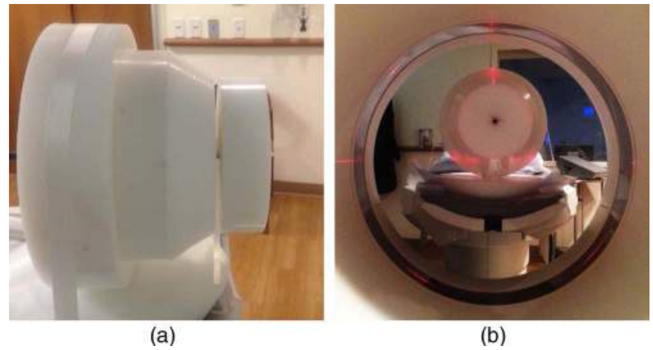

FIG. 5. Pictures of the experimental setup of the proposed technique. (a) In the first step, the scanning section of the phantom was extended beyond the patient table to avoid the influence of the patient bed in the followed CT scan. (b) In the second step, the phantom was positioned off-center in the SFOV by adjusting vertically the patient bed.

To reduce the influence of the scatter radiation, a 2×1 mm collimation was used. A Monte Carlo simulation with MC-GPU (Ref. 30) indicated that the scatter-primary ratios (SPRs) of position 1 in Fig. 3 were less than 0.008 for 140 kVp, consistent with the results of previous studies (Fig. 6).[31,32] As the SPRs of the transmission data were further reduced by the built-in scatter correction strategies of Somatom Flash Definition scanner[26] (e.g., an antiscatter grid), we postulated that the actual SPRs of the transmission data in our study should be less than that value estimated by the Monte Carlo simulation, and then inconsequential to the measurements. We further demonstrated in a secondary analysis that the distance between the phantom and the detector had negligible influence on the transmission measurements (see Fig. 9).

The maximum fan angle of the estimated spectra depends on the phantom radius r and the distance d between the phantom center and the rotation center. If the x-ray path is required to pass through the phantom center to achieve the maximum intersection length of $2r$ (Fig. 7), the maximum fan angle would be

$$\varphi = \pm \frac{180^\circ}{\pi} \arcsin\left(\frac{d}{\mathrm{SOD}}\right), \tag{16}$$

where SOD is the source-to-object distance (i.e., 595 mm). In that condition, based on our experimental setup ($d = 80$ mm, i.e., position 1 in Fig. 3), the angular trajectories would be within $[\theta_{114}, \theta_{622}] = [-8^\circ, 8^\circ]$, where the subscript in θ_n indicates the detector index. However, our results (in Table IV) indicate that it is possible to obtain spectra with good accuracy even if the maximum intersection length is reduced to r

TABLE II. Technical description of the Al and Cu filters at 140 kVp.

Material	Density (g/cm^3)		1	2	3	4	5	6	7	8	9
Al	2.700	L (mm)	1.000	3.000	5.000	7.500	10.500	14.500	20.500	25.500	40.500
		Transmission $Y(L)$	0.936	0.819	0.721	0.616	0.514	0.401	0.286	0.206	0.098
Cu	8.960	L (mm)	0.127	0.254	0.655	1.062	1.562	2.090	3.124	3.658	4.686
		Transmission $Y(L)$	0.864	0.761	0.546	0.415	0.303	0.229	0.140	0.112	0.074

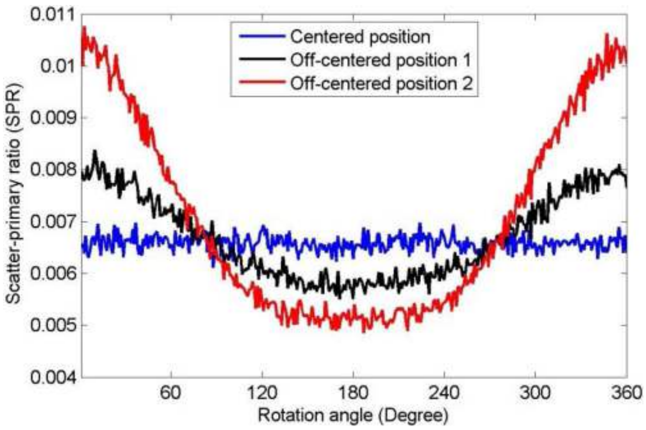

FIG. 6. Monte Carlo simulation results of the maximum scatter primary ratio as a function of rotation degree for three different phantom positions, i.e., the centered position, the off-centered position 1 in Fig. 3, and the off-centered position 2 in Fig. 3. The geometry of the Somatom Definition Flash scanner was simulated with a perfect detector (2 × 1 mm collimation) and without an antiscatter grid using Monte Carlo simulation code MC-GPU (Ref. 30). A cylindrical phantom made of polyethylene (160 mm in diameter and 250 mm in height) was imaged at 140 kVp (Ref. 31).

(Fig. 7). In that condition, the transmission measurements are reduced to about 0.2 and the maximum fan angle is extended to

$$\varphi' = \pm\frac{180^\circ}{\pi}\arcsin\left(\frac{d+\sqrt{3}r/2}{\mathrm{SOD}}\right). \tag{17}$$

According to the above equation, the range of the angular trajectories was extended to $[\theta_{147},\theta_{589}] = [-15^\circ,\ 15^\circ]$ for the geometry used in this experiment. The coverage of these results is less than the CT scanner's fan angle (i.e., $[-25^\circ, 25^\circ]$). The quarter detector offset was considered negligible assuming the angle of the central detector element to be zero, i.e., $\theta_{368} = 0^\circ$.

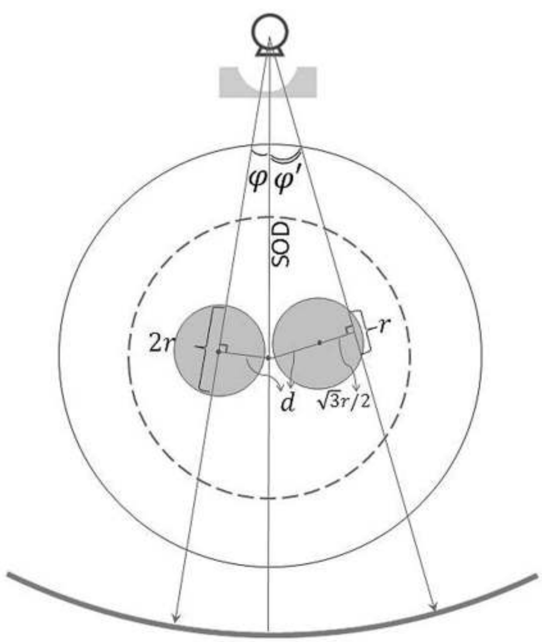

FIG. 7. Schematic diagram to illustrate the maximum fan angles φ and φ' determined by the maximum intersection lengths of $2r$ and r, respectively.

The incident spectrum of the central detector element at 140 kVp was estimated by the conventional and the proposed technique. The results were compared in terms of the mean energy difference (MED) and the normalized root mean square deviation (NRMSD). The MED was defined as the mean energy of the estimated spectrum minus the mean energy of the manufacturer-provided spectrum. The NRMSD was defined as

$$\mathrm{NRMSD} = \frac{\sqrt{\sum_s (N_s - N_s^*)^2/N_E}}{\max(N_s^*) - \min(N_s^*)}, \tag{18}$$

where N_s^* is the expected spectrum, N_s is the estimated spectrum, and N_E is the total number of the energy bins. In addition, the incident spectra of the detector elements between -15° and 15° were estimated and were quantitatively compared with the incident spectra derived from the manufacturer-provided spectra. Note that as the manufacturer-provided spectra cannot be assumed to be the actual gold standard, the MED and NRMSD may not be interpreted as metrics of absolute accuracy.

3. RESULTS

3.A. The conventional technique vs. the proposed technique

The transmission measurement results for the central detector element ($\theta_{368} = 0^\circ$) at 140 kVp are plotted in Fig. 8. Figure 8(a) shows the results of the metal filters (i.e., Al and Cu) using the conventional technique and Fig. 8(b) shows the results of the polyethylene phantom using the proposed technique. Due to the time consuming nature of the conventional technique, only a single-kVp spot-validation at 140 kVp was employed. Both of the experimental results in Figs. 8(a) and 8(b) yielded a good agreement with the simulation results (solid lines). However, due to the limited number of metal filters, Fig. 8(a) included fewer measurements. In contrast, the latter figure included more than 900 measurements, to which a high-degree polynomial function could be fitted to reduce the influence of noise and to more accurately represent the transmission measurements.

Figure 9 shows the transmission measurements acquired by the central detector at different locations. The circles and dots indicate the data acquired when the phantom was near to the x-ray tube and the detector, respectively. Due to the built-in scatter reduction techniques (e.g., antiscatter grid) and the narrow collimation, the influence of the scatter radiation on the measurements could be reduced to negligible levels; both datasets were close to the simulated data (solid line) with no notable difference. The fluctuation of the measured transmission measurements was mainly due to the quantum noise and the finite size of the detector (0.6 × 0.6 mm^2). The latter factor could impact the accuracy of the derived intersection lengths. With our data conditioning procedure, the resampled measurements (solid squares) had a good agreement with the

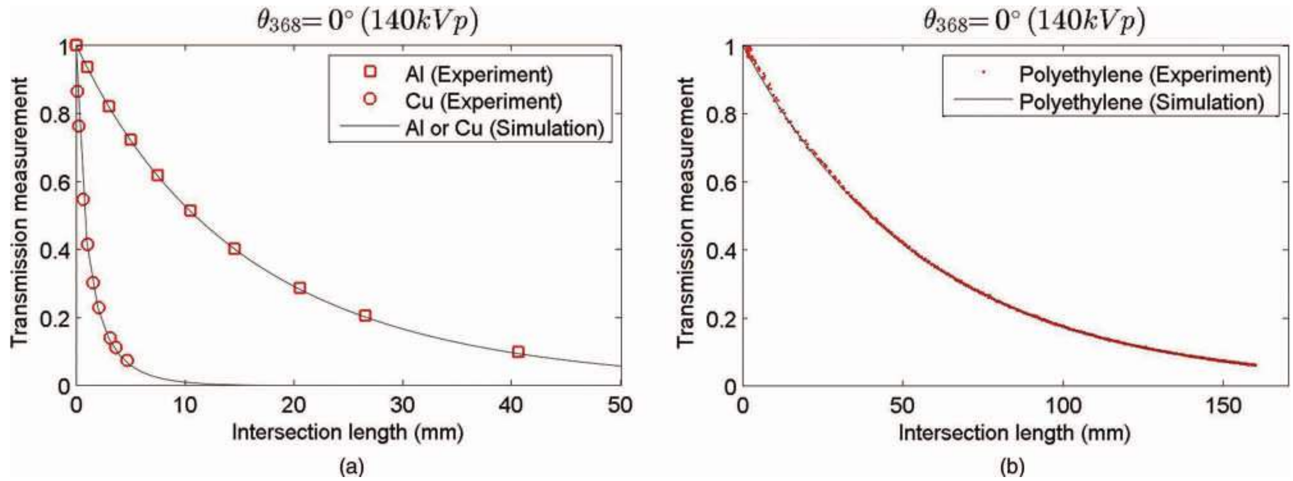

FIG. 8. Comparison of the experimental results and the simulated results in terms of the transmission measurement as a function of the intersection length for the central detector element ($\theta_{368} = 0°$) at 140 kVp. (a) Experimental measurement results of Al and Cu using the conventional technique. (b) Experimental measurement results of polyethylene using the proposed technique. The estimated error for polyethylene (too small to be visualized) was 0.002, determined by computing the standard deviation of the transmission measurements within the 1 mm interval centered at the intersection length of 137 mm.

simulated measurements and the fluctuation was significantly suppressed.

Using the corresponding transmission measurement (Fig. 8), the incident spectrum of the central detector element was estimated and compared with the initial spectrum and the manufacturer-provided spectrum. The results show that the proposed technique [Fig. 10(b)] yielded spectral estimation comparable to those of the conventional technique [Fig. 10(a)]. Detailed quantitative comparison in terms of MED and NRMSD (Table III) shows a slightly closer correlation for our results with the manufacturer-provided spectrum.

It is notable that the NRMSD of the conventional technique here was 1.56%, which was much smaller than that in Duan's work[18] (i.e., [4.7%, 8.6%]). This improvement might be due to the use of more reliable reference spectra in this work, which were provided by the manufacturer instead of being simulated with software. The MED value

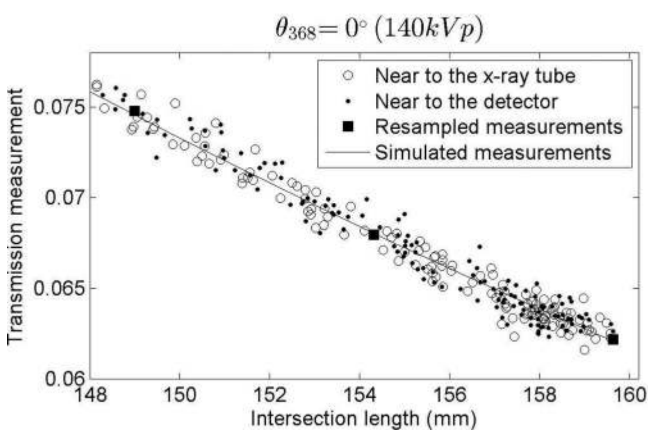

FIG. 9. Comparison of the transmission measurements acquired by the central detector when the phantom was near to the x-ray tube (circles) and near to the detector (dots). The resampled measurements after data conditioning procedure were plotted as solid squares. The simulated measurements were plotted as a solid curve.

(0.64 keV) was slightly larger than those reported in Duan's work[18] (i.e., [−0.23, 0.55]), but still within the statistical uncertainty.

3.B. Incident spectra of different angular trajectories

With the proposed technique, the incident spectra for four noncentral x-ray trajectories (i.e., $\theta_{324} = -3°$, $\theta_{265} = -7°$, $\theta_{206} = -11°$, $\theta_{147} = -15°$) were estimated and are plotted in Fig. 11. The reference spectra were derived from the manufacturer-provided spectrum and the bowtie filter. The comparative results show that the estimated spectra accorded well with the reference spectra across different trajectories. The quantitative results in terms of MED and NRMSD are shown in Table IV. The results of the extended angular trajectories (i.e., θ_{206} and θ_{147}) were within the statistical uncertainty of the results of the nonextended angular trajectories (i.e., θ_{324} and θ_{265}).

In addition, the beam hardening effect could be observed in these figures, as the photon distribution gradually shifted from low energy to high energy with the increase of the absolute value of the beam angle. This phenomenon can be better appreciated in Fig. 12, where the incident spectra across a wide range of angular trajectories ($\theta_n \in [\theta_{147}, \theta_{368}] = [-15°, 0°]$) were estimated by the proposed technique. From Figs. 12(a) to 12(d), different kVp settings (i.e., 140, 120, 100, and 80 kVp) were used.

With the estimated spectra, their corresponding mean energies were evaluated as a function of beam angle (Fig. 13). The mean energies derived from the proposed technique (dots) had an excellent agreement with the mean energies derived from the manufacturer-provided prebowtie spectra (solid lines).

The estimation performance of different kVp settings was also assessed (Table V). The averaged absolute MEDs across the beam angles between [−15°, 15°] of all x-ray tube spectra were less than 0.61 keV with standard deviation less than

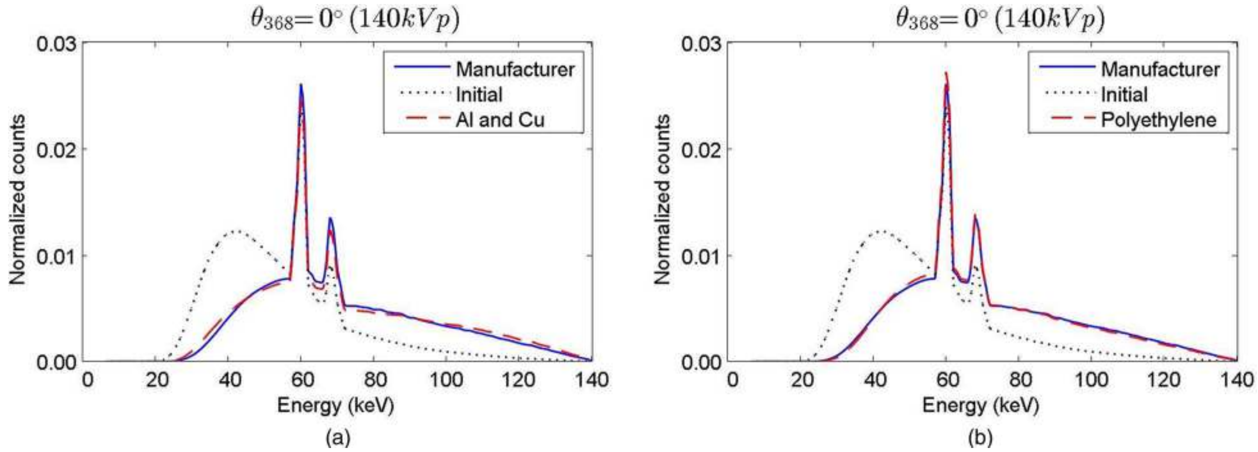

FIG. 10. Comparison of the central spectrum derived from the manufacturer-provided prebowtie spectrum (solid line), the central spectrum estimated with Al and Cu using the conventional technique [dashed line in (a)], and the central spectrum estimated with polyethylene using the proposed technique [dashed line in (b)]. The same initial spectrum (dotted line) was used by the two techniques.

0.74 keV. The averaged NRMSDs were less than 3.41% with standard deviation less than 2.02%. These quantities indicate the high estimation accuracy and stability of the proposed technique across different angular trajectories and kVp settings.

With the decrease of the kVp, the averaged NRMSD increased. This is likely due to the decrease in the maximum number of photons in the spectrum, [i.e., $\max(N_s^*) = 0.026$, 0.025, 0.022, 0.016 for 140, 120, 100, 80 kVp, respectively], which magnified the NRMSD values via the normalization process in Eq. (18). In addition, with decreased kVp, more photons were attenuated leading to increased contribution of the quantum noise and scatter radiation.[31,32]

It was observed that the shape of the curves (Fig. 13) had an angle-dependent pattern at higher kVp settings (i.e., 140, 120, and 100 kVp). The mean energies along certain angular trajectories were systematically smaller (e.g., $-8°$ and $8°$) or larger (e.g., $-6°$ and $-1°$) than the simulated mean energies. These patterns were probably caused by systematic errors.

4. DISCUSSION

In this work, an angle-dependent estimation technique of CT x-ray spectrum from rotational transmission measurements was presented. This method enables the estimation of the incident spectra across a wide range of angular trajectories with a single phantom and a single axial CT scan in the absence of the knowledge of the bowtie filter. Compared with the previous techniques, the proposed technique has the following two important features: (1) estimation of the spectra with an off-centered geometry and (2) an effective data conditioning procedure. With an off-centered geometry, transmission measurements along different angular trajectories can be collected with one rotation using the CT detector elements. This acquisition geometry has three merits. First, the CT scanner can be operated in axial mode instead of in service mode, which is not always accessible in all CT scanners. Second, the axial mode scan greatly increases the data collection efficiency, as transmission measurements across a wide range of angular trajectories can be quasicontinuously collected within a short time. Finally, the collected measurements can be used to reconstruct the cross sectional image of the phantom to facilitate deriving the intersection lengths.

Because of the limited number of the measurements, the previous techniques do not have the data conditioning procedure. Therefore, diverse attenuation materials have to be adopted to reduce noise and to improve estimation stability. In comparison, the proposed technique can collect hundreds of transmission measurements for each angular trajectory. With the abundant transmission measurements, polynomial functions with the correct boundary condition can be used to fit the data, which helps reduce the influence of the dramatic change of the intersection lengths derived from the x-ray paths near the edge of the phantom. Therefore, this data conditioning procedure reduces noise and improves both the estimation accuracy and the estimation stability.

An experiment was conducted by comparing the conventional technique and the proposed technique. The results show that, for the incident spectrum of the central beam, the MED and the NRMSD of the proposed technique were both largely comparable to those of the conventional technique. The incident spectra across a wide range of angular trajectories were also estimated. The quantitative results show that the average absolute MED and averaged NRMSD of all kVp settings were smaller than 0.61 keV and 3.41%, respectively,

TABLE III. Comparison of the two techniques in terms of MED and NRMSD for the incident spectrum of 140 kVp.

Beam angle	Method	MED (keV)	NRMSD (%)
$\theta_{368} = 0°$	Conventional technique	0.64	1.56
	Proposed technique	−0.28	0.84

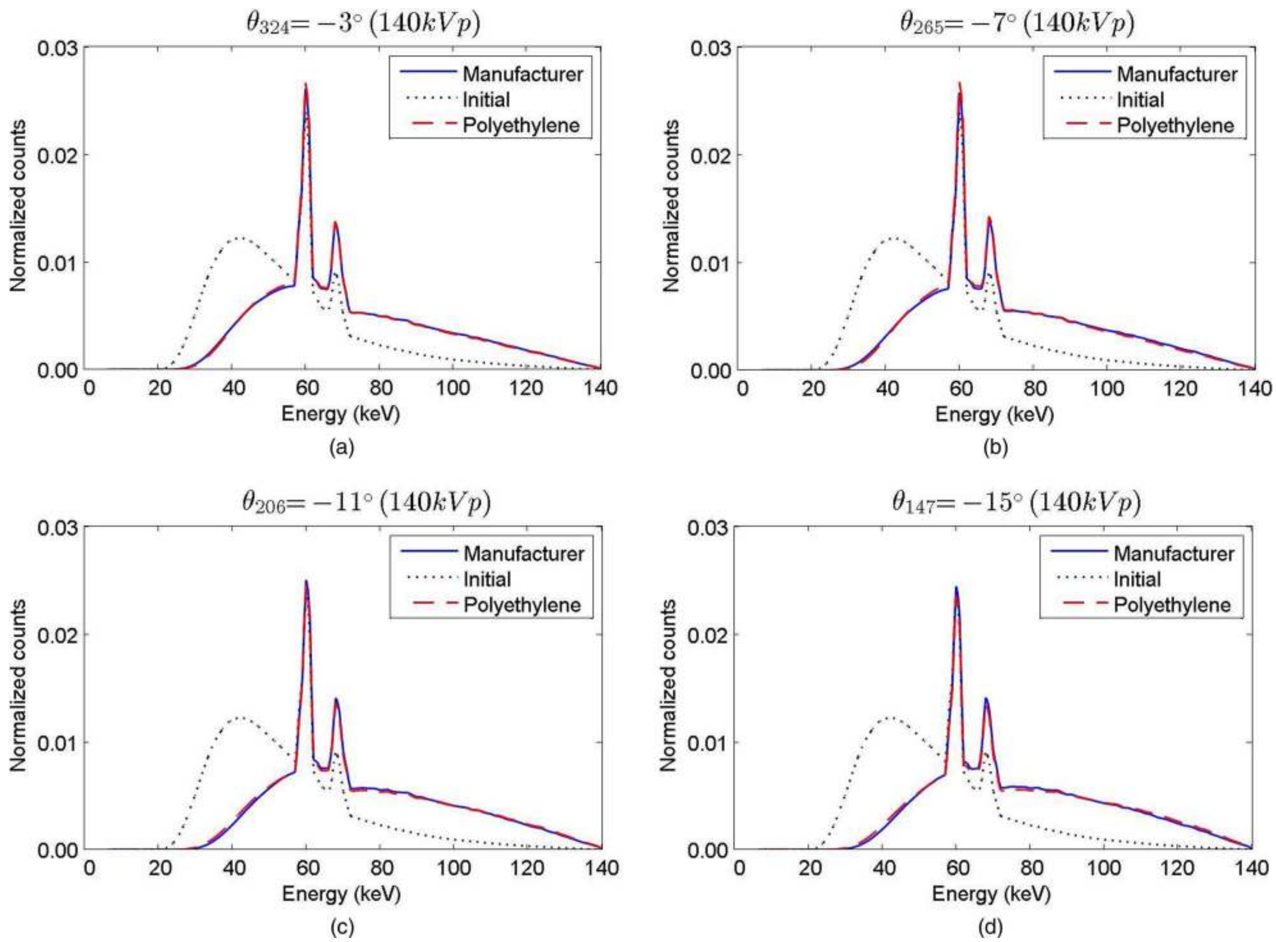

FIG. 11. Comparison of the spectra derived from the manufacturer-provided prebowtie spectra (solid lines) and the spectra estimated with polyethylene (dashed lines) from different beam angles [(a) $\theta_{324} = -3^\circ$, (b) $\theta_{265} = -7^\circ$, (c) $\theta_{206} = -11^\circ$, and (d) $\theta_{147} = -15^\circ$]. The same initial spectrum (dotted line) was used.

which indicates the high accuracy and stability of the proposed technique across different angular trajectories and kVp settings.

Two types of systematic errors might contribute to the angle-dependent pattern in Fig. 13. The first one is the inherent systematic errors of the detector modules[33,34] (e.g., gain error, nonlinearity error, and energy sensitivity difference, etc.), as it is impossible to produce identical detector modules in the manufacturing process. An additional possible cause might be due to the unexpected deviation of the bowtie filters. Other possible sources of error remain topics for future research.

TABLE IV. Comparison of the estimated spectra from different beam angles in terms of MED and NRMSD for the incident spectrum of 140 kVp.

Beam angle	Maximum intersection length (mm)	MED (keV)	NRMSD (%)
$\theta_{324} = -3^\circ$	160	−0.20	0.45
$\theta_{265} = -7^\circ$	160	−0.49	0.76
$\theta_{206} = -11^\circ$	148	−0.11	0.68
$\theta_{147} = -15^\circ$	72	−0.27	0.93

In this work, the manufacturer-provided spectra were used as reference spectra, which neglect the possible spectral characteristics of each individual x-ray system and thus cannot be used as a gold standard. Future studies may include validation experiments (e.g., using the transmission measurements with a different material or more materials) to further quantify the accuracy and stability of the proposed method.

There are several ways by which the method developed in this study can be improved. First, one of the major sources of error in our method is the limited resolution of the detector. This error can be reduced using the flying focal spot technique (FFS),[35] which can improve the resolution from 0.6×0.6 to 0.3×0.3 mm^2. Second, while we use a high mAs setting and a data conditioning procedure to reduce the quantum noise, it is possible to further reduce the quantum noise by averaging multiple axial scans. Third, if we use position 2 in Fig. 3, the range of the angular trajectories can be extended to $\theta_n \in [\theta_{17}, \theta_{719}] = [-24^\circ, 24^\circ]$, which almost covers the whole detector range, i.e., $[-25^\circ, 25^\circ]$. This extension however requires certain accommodations. In order to avoid the influence of the scatter radiation for this position, we can only utilize the transmission measurements acquired near to the x-ray tube for spectral estimation. Fourth, it is straightforward to extend the proposed method to estimate the spectra along the

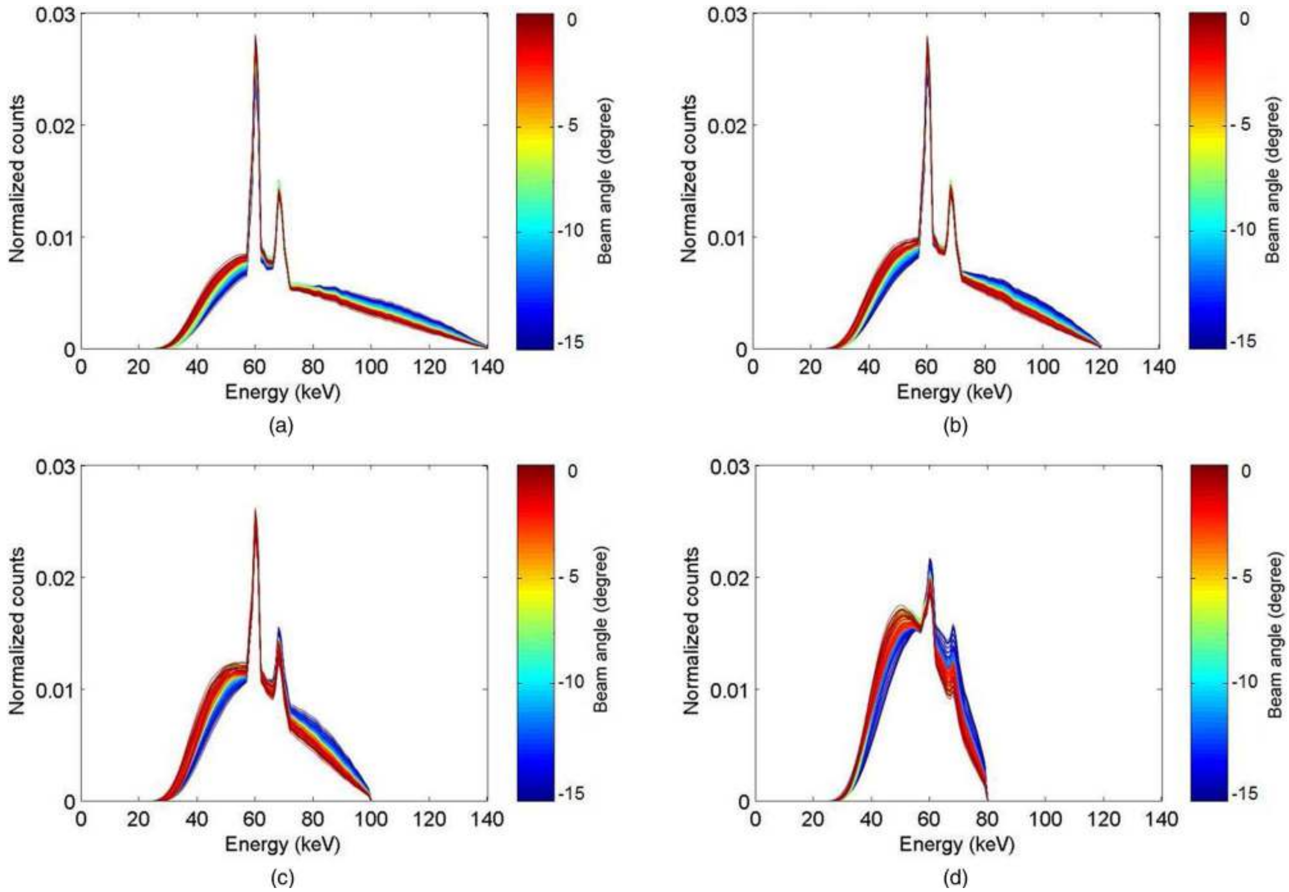

FIG. 12. Plot of the estimated spectra from different beam angles ($\theta_n \in [\theta_{147}, \theta_{368}] = [-15^\circ, 0^\circ]$) with different kVp settings, i.e., (a) 140, (b), 120, (c) 100, and (d) 80 kVp. The jet color map was used to differentiate spectra along different angular trajectories.

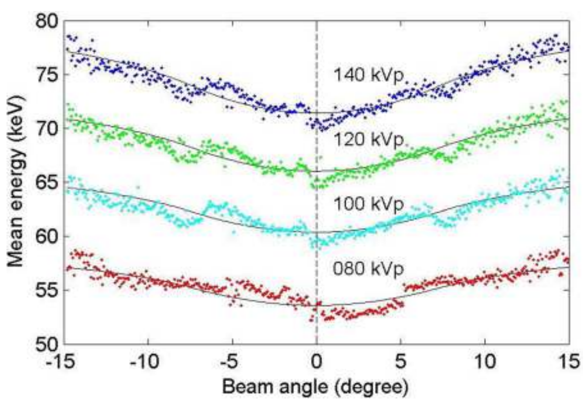

FIG. 13. Plots of the mean energy of the incident spectrum as a function of the beam angle ($\theta_n \in [\theta_{147}, \theta_{589}] = [-15^\circ, 15^\circ]$). The solid lines were derived from the manufacturer-provided spectra; the dots were derived from the proposed technique. From top to bottom, the results were derived from different kVp settings, i.e., 140, 120, 100, and 80 kVp.

cone-angle (i.e., z direction). However, with the increase of the cone angle, the scatter radiation will also increase and can potentially contaminate the transmission measurements and result in unreliable estimated spectra. If the knowledge of the bowtie filter (i.e., the attenuation properties and the geometry information of the bowtie filter) is available, spectra can be estimated with a narrow fan-beam collimation and then used to derive the spectra along the cone-angle with the knowledge of the bowtie filter. Furthermore, the knowledge of the bowtie filter can be incorporated into the system matrix [Eq. (3)], such that all measurements can be used to reconstruct a single prebowtie spectrum. That spectrum can be used to derive the spectra along any angular trajectory. As this technique uses more data and a prior known knowledge of the bowtie filter, it can potentially yield more precise estimation of the spectra. These possible extensions of the work will remain topics for future investigation.

TABLE V. Quantitative analysis in terms of the averaged absolute MED, the standard deviation of the MEDs, the averaged NRMSD, and standard deviation of the NRMSDs across the beam angles between $[-15^\circ, 15^\circ]$ for different kVp settings.

Beam angle	Spectrum (kVp)	Averaged absolute MED (keV)	Standard deviation of the MEDs (keV)	Averaged NRMSD (%)	Standard deviation of the NRMSDs (%)
$[\theta_{147}, \theta_{589}] = [-15^\circ, 15^\circ]$	140	0.53	0.66	0.89	0.64
	120	0.57	0.69	1.08	0.75
	100	0.59	0.71	1.60	1.04
	80	0.61	0.74	3.41	2.02

5. CONCLUSIONS

In this work, we proposed a novel angle-dependent estimation technique of CT x-ray spectrum from rotational transmission measurements. In comparison with the conventional techniques, the proposed technique simplifies the measurement procedures, enables the incident spectral estimation ability for a wide range of angular trajectories, and improves the estimation accuracy and stability. This technique can be applied to both rigorous research objectives (e.g., dose estimation and image reconstruction) and routine clinical quality control procedures.

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